

RICHA REDDY BADALA

AI/ML Engineer

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SUMMARY

Machine Learning Engineer with 5+ years of experience building production-grade AI/ML and GenAI systems, including LLM-powered copilots and real-time decisioning platforms. Specialized in Retrieval-Augmented Generation (RAG), distributed ML systems, and low-latency inference at scale using Kafka, Spark, and Kubernetes. Proven track record of driving measurable impact—improving productivity, reducing latency, and increasing conversion and fraud detection effectiveness. Strong expertise in MLOps, cloud-native architectures (AWS/Azure), and secure, enterprise-grade AI deployment.

PROFESSIONAL EXPERIENCE

JP Morgan Chase

Feb 2024 – Present

Machine Learning Engineer

San Francisco, CA

- Built and scaled a GenAI-powered developer copilot using LLM + RAG pipelines, improving engineering productivity by 25% across internal teams.
- Designed a real-time context-aware code generation system integrating enterprise codebases and APIs, achieving 85%+ suggestion acceptance rate.
- Implemented caching and latency optimization strategies using Redis and prompt compression techniques, reducing end-to-end response latency by 40% for high-concurrency workloads.
- Implemented secure LLM guardrails and fine-grained data access controls, enabling safe GenAI adoption across thousands of developers while ensuring compliance with enterprise security policies.
- Developed end-to-end RAG pipelines using LangChain, FAISS/Pinecone, and OpenAI embeddings to retrieve and rank internal code, documentation, and API schemas for context-aware responses.
- Built scalable LLM inference services using Python (FastAPI, async processing, gRPC) deployed on Kubernetes (EKS/AKS) with autoscaling and GPU-backed workloads for low-latency serving.
- Engineered real-time streaming pipelines using Apache Kafka and Spark Structured Streaming to process code updates, logs, and metadata for near real-time embedding refresh and synchronization.
- Designed cloud-native ML infrastructure leveraging AWS (SageMaker, Bedrock, S3, Lambda) and Azure OpenAI with secure VPC endpoints, IAM-based access control, and encrypted data pipelines.
- Built AI evaluation and monitoring pipelines using MLflow, Prometheus, and Datadog to track model performance, hallucination rates, and system usage, enabling continuous improvement via feedback loops.

IBM

July 2020 – Nov 2022

Machine Learning Engineer

India

- Built and deployed a real-time fraud detection and transaction scoring system processing millions of daily events, reducing fraudulent transactions by 32%
- Designed an AI-driven personalization and AdTech monetization engine (Next Best Offer + CTR models), increasing in-app offer conversion rates by 27%
- Engineered a low-latency decisioning platform (<100ms SLA) for payments and recommendations, improving digital user engagement by 22%
- Developed event-driven ML pipelines using Apache Kafka, Spark Structured Streaming, and Python (PySpark) to process real-time transaction and user behavior data for fraud detection and personalization
- Built and deployed scalable ML inference services using FastAPI, Docker, and Kubernetes (OpenShift/EKS), enabling high-throughput, low-latency model serving across microservices
- Designed feature engineering and feature store pipelines using Feast, Redis, and Cassandra, supporting both batch and real-time ML use cases with consistent feature availability
- Implemented end-to-end MLOps workflows using MLflow, Airflow, and GitHub Actions, enabling automated model training, versioning, CI/CD deployment, and monitoring in production
- Developed AI-powered document and media processing pipelines (OCR + NLP + computer vision) using Tesseract, OpenCV, and transformer-based models, automating KYC verification and reducing onboarding friction

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL

AI / Machine Learning: Machine Learning, Deep Learning, NLP, Recommendation Systems, Fraud Detection, Personalization Models, CTR Prediction, Time-Series Modeling

Generative AI / LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Prompt Compression, LLM Guardrails & Safety, Context-Aware Code Generation, AI Copilots, Embedding Models, Semantic Search, Hallucination Detection & Mitigation

AI Frameworks & Tools: LangChain, OpenAI API, Azure OpenAI, Amazon Bedrock, Hugging Face Transformers, FAISS, Pinecone, Tesseract OCR, OpenCV

Data Engineering & Streaming: Apache Kafka, Spark Structured Streaming, PySpark, Real-Time Data Pipelines, Data Processing at Scale

Backend & APIs: FastAPI, gRPC, REST APIs

Architecture & Distributed Systems: Event-Driven Architecture, Async Processing, Microservices Architecture, High-Concurrency Systems, Low-Latency System Design, Cloud-Native Architecture

MLOps & Model Lifecycle: MLflow, Airflow, GitHub Actions, CI/CD for ML, Model Monitoring, Model Evaluation Pipelines, Feature Stores (Feast), Experiment Tracking, A/B Testing

Cloud Platforms: AWS (SageMaker, Bedrock, S3, Lambda, EKS, IAM, VPC), Microsoft Azure (Azure OpenAI, AKS)

Databases & Storage: Redis, Cassandra, Vector Databases, Data Lakes

DevOps & Infrastructure: Docker, Kubernetes, GPU Workloads, Autoscaling, Infrastructure as Code (IaC)
Monitoring & Observability: Prometheus, Datadog, Logging, Metrics, Alerting, LLM Evaluation & Usage Analytics

Security & Compliance: Secure AI Systems, Data Access Control, IAM, Encryption, Enterprise AI Governance, LLM Safety & Guardrails

EDUCATION

Roosevelt University

Master of Science in Computer Science

CERTIFICATIONS

AWS Certified Solutions Architect - Associate